

PE-RankFormer: A Minimally-Mechanistic Relational Transformer for Prime-Editing Efficiency Prediction

PEFormer research pilot

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Abstract

Prime editing (PE) efficiency prediction has been dominated by mechanistic models that hard-code assumptions about the PE reaction pathway (nick, flap resolution, mismatch repair, etc.), most recently OptiPrime (Hsu et al., 2026), a strongly mechanistic model trained on 297,962 PE experiments. We ask whether a minimally-mechanistic, general-purpose relational Transformer, given the same scale of training data, can match or exceed it. We reconstruct 262,508 of the 297,962 target training examples (88.1%) from four source studies, implement PE-RankFormer – a 26.6M-parameter paired-sequence Transformer with cross-attention between the edit-outcome encoding and the pegRNA encoding and FiLM-based conditioning on experimental context – and train it with a joint regression and within-target ranking objective. A single architectural addition motivated by diagnosing early overfitting, a 3-way categorical “simplex” outcome head that models the mutually-exclusive unedited/edited/indel read fractions instead of regressing edited-read fraction alone, produced the largest single gain in the project. A 6-model ensemble of PE-RankFormer variants, evaluated head-to-head against OptiPrime’s own released 5-model ensemble on the 15,022-row Hsu subset of the locked test fold, achieves Spearman $\rho = 0.807$ versus OptiPrime’s 0.724 (Pearson 0.780 vs. 0.724), a difference of +0.083 with a protospacer-clustered paired bootstrap 95% CI of [+0.047, +0.118] ($p < 0.0005$, PE-RankFormer wins in 2000/2000 resamples). We report this result together with negative results from the same sweep and explicit caveats about corpus completeness and test-fold query discipline.

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1 Introduction

Prime editing efficiency prediction is typically approached with strongly mechanistic models: pipelines that explicitly encode the PE reaction as a sequence of biochemical steps (spacer/PBS/RTT annealing, flap competition, mismatch-repair outcome) and use that structure to constrain the model. OptiPrime (Hsu et al., 2026) is the current state of the art on this axis, trained on 297,962 PE experiments assembled across four studies.

The research question motivating this pilot is whether a *minimally mechanistic* model – one that encodes only the raw sequences and experimental metadata and lets a general-purpose relational architecture (a Transformer with cross-attention between the edit and pegRNA representations) discover whatever structure is predictive – can match or outperform a strongly mechanistic model of comparable training scale. This is a real test of an increasingly common hypothesis in computational biology: that architectural inductive bias from domain mechanism can be substituted for by scale plus a sufficiently expressive general-purpose architecture, provided the objective is designed to exploit the structure actually present in the data (e.g., within-target ranking, or – as turned out to matter most in this pilot – the full space of mutually exclusive sequencing outcomes rather than a single scalar summary of them).

This report documents the full pipeline: corpus reconstruction, architecture, training procedure, a systematic ablation and configuration sweep, and a statistically validated, ensemble-matched head-to-head comparison against OptiPrime’s own released baseline.

2 Training Corpus Reconstruction

OptiPrime was trained on 297,962 PE experiments assembled from four source studies. We reconstructed this corpus independently from public data releases, verified against a 42-partition ground-truth breakdown (by study $\times$ cell type $\times$ PE system) reported in the OptiPrime paper.

Table 1: Final reconstructed corpus composition (`data/processed/optiprime_full_297962.parquet`).

Source	Partitions matched (of target)	Rows
Liu (Hsu 2026, newly generated)	4/4 (raw count; see note)	74,769
Kim (DeepPrime)	18/18 exact	58,301
Schwank (PRIDICT / PRIDICT2.0)	18/20 exact	129,438
Total	36/42 target-exact	262,508

262,508 of 297,962 target rows (88.1%) were recovered with a documented, reproducible provenance trail. The three remaining discrepancies were investigated and ruled out rather than papered over: (1) the Hsu/Liu partition contains 74,769 raw rows against a 65,594-row target with no public QC/coverage column to explain the gap – several plausible filters (PE2/PE4 co-presence, barcode-collision exclusion, invalid-read exclusion, zero-efficiency exclusion, OptiPrime’s own embedding-size bounds) were each tested and rejected, so we use the full 74,769; (2) the Schwank K562 PE4+epegRNA partition (target 21,201) is essentially absent from any public release (823 rows found, after checking PRIDICT v1’s full supplementary branch, PRIDICT2.0’s processed data, and the ranking-percentile file); (3) the Schwank K562 PE2+epegRNA partition

has a small 94-row excess traced to an unapplied PRIDICT2.0 QC exclusion list. Full detail in `reports/training_data_reconstruction.md` and `reports/dataset_reconstruction_status.md`.

Cross-validation folds were built to be **protospacer-disjoint**: fold 0 (test), fold 1 (val), folds 2–4 (train), seed 20260812, with zero-leakage verified (no protospacer appears in more than one fold).

3 Model Architecture

PE-RankFormer is a 26.6M-parameter Transformer ($d_{\text{model}}=384$, 6 attention heads, FFN width 1536, dropout 0.10) with two parallel token encoders and a bidirectional cross-attention stage:

- **Edit encoder** (6 layers): a paired-sequence encoding of the unedited and edited full target sequence, so the model can directly attend to what changed rather than being told a structured description of the edit.
- **PegRNA encoder** (4 layers): a segment-aware encoding of the spacer, PBS, and RTT, with segment-type embeddings distinguishing the three regions.
- **Cross-attention** (2 blocks): bidirectional cross-attention between the edit-token and pegRNA-token representations, followed by attention pooling on each side.
- **FiLM conditioning**: pooled representations are modulated by an embedding of experimental context (cell type, PE system variant, Cas9/PAM variant, scaffold, motif), so the same sequence pair can receive different predictions in different experimental contexts.
- **Prediction head**: either a scalar sigmoid head (edited-fraction regression) or the 3-way simplex head described in §3.1.

No PE-reaction-specific mechanism (e.g., explicit flap-competition or mismatch-repair modeling) is encoded; all such structure, if present in the data, must be discovered by the cross-attention mechanism itself – this is the “minimally mechanistic” design constraint against which the model is being tested.

3.1 The simplex outcome head: diagnosis-driven innovation

Early training runs (Model B, scalar head, no ranking loss) showed clear overfitting: training loss fell roughly $4\times$ over 30 epochs while validation Spearman ρ plateaued from epoch ~ 20 onward (Figure 1). Adding model capacity would not address this; the constraint was in the supervision signal, not the architecture’s expressive power.

Every PE sequencing read falls into exactly one of three mutually exclusive categories: *unedited*, *edited* (the intended outcome), or *indel* (an unintended byproduct of the nick/repair process). The original scalar-head design discarded the indel read fraction entirely and regressed only the edited fraction – despite indel being present with 100% coverage in the reconstructed corpus, nonzero in $\sim 60\%$ of rows, and correlated with editing efficiency ($r \approx 0.25$). This is a real, cheaply available signal being thrown away.

We replaced the scalar sigmoid head with a **3-way simplex head**: a softmax over (unedited, edited, indel) logits trained with soft cross-entropy against the observed read proportions (clipped/renormalized on the rare rows where edited+indel exceeds 1 due to counting noise). At inference, $\text{efficiency} = \text{softmax}(\text{logits})_{\text{edited}}$, and the ranking score used for the pairwise ranking loss is the edited-vs-unedited log-odds, $\text{logit}_{\text{edited}} - \text{logit}_{\text{unedited}}$.

This does not violate the minimally-mechanistic constraint: mutual exclusivity of the three read categories is a property of how sequencing reads are classified (an assay-level fact), not an assumption about the underlying PE reaction graph. No biochemical step ordering or intermediate state is encoded.

The simplex head produced the top 4 single-model validation Spearman results of the entire sweep (§5). Two further variations, tried to see whether the gain could be pushed further, produced negative results and are reported as such rather than discarded: training the regression component in clipped-logit space instead of raw $[0, 1]$ space gave no improvement (0.866 vs. 0.864 raw, within seed noise), and increasing dropout to 0.20 gave no improvement (0.871 vs. 0.870 at dropout 0.10).

4 Training Procedure

Models are trained with joint regression (Huber loss) and pairwise RankNet within-target ranking loss (λ_{rank} weighting), using a custom grouped batch sampler that biases minibatches toward rows sharing a within-target ranking group – necessary because the $\sim 220,000$ distinct ranking groups in the full corpus mean uniform random batching yields close to zero rankable pairs per batch. Optimization uses AdamW with a warmup + cosine schedule, BF16 mixed precision, and fused scaled-dot-product attention, on a single RTX PRO 6000 Blackwell GPU. All models in the final sweep were trained for a full 30 epochs (early-stopping patience matched across configurations after an initial unfair comparison was caught and corrected – see §7).

5 Ablation and Configuration Sweep

A bounded sweep over outcome head (scalar vs. simplex), regression target space (raw $[0, 1]$ vs. clipped-logit), ranking-loss weight λ_{rank} , context conditioning (on/off), and dropout was run, each configuration trained to full 30 epochs at matched patience. All selection decisions in this sweep were made on the **validation** fold only.

Table 2: Full configuration sweep, sorted by best validation Spearman ρ (single seed 20260812 unless noted). “rank” = with ranking loss ($\lambda_{\text{rank}}=0.25$ for Models A/C, 0.05 for the low-rank probe); blank = ranking loss off ($\lambda_{\text{rank}}=0$).

Run	Head	Space	Context	λ_{rank}		Val ρ
final_simplex_s2	simplex	raw	on	0.00		0.879
final_simplex_s3	simplex	raw	on	0.00		0.873
exp_dropout20_simplex	simplex	raw	on	0.00	0.871 (dropout 0.20)	
exp_simplex	simplex	raw	on	0.00		0.870
model_b_norank	scalar	raw	on	0.00		0.864
final_logit_s3	scalar	logit	on	0.00		0.867
exp_logit_norank	scalar	logit	on	0.00		0.866
final_logit_s2	scalar	logit	on	0.00		0.862
exp_lowrank005	scalar	raw	on	0.05		0.850
model_a_rank	scalar	raw	on	0.25		0.786
model_c_nocontext_fair	scalar	raw	off	0.25		0.705
model_c_nocontext	scalar	raw	off	0.25	0.675 (early-stopped, unfair)	

Three findings stand out. First, the simplex head is the single largest lever in the sweep, outperforming every scalar-head configuration regardless of other settings. Second, the ranking loss *hurts* single-model validation Spearman at the weights tried ($\lambda_{\text{rank}}=0.25$ costs $\sim 0.08 \rho$ relative to $\lambda_{\text{rank}}=0$); this is a genuine trade-off against within-target ranking quality that a scalar global Spearman number does not fully capture (see §6.2 for within-target metrics), and is noted as a caveat rather than resolved in this pilot. Third, removing experimental-context conditioning is unambiguously costly (0.705 vs. 0.786 at matched patience, both with ranking loss on), confirming the FiLM conditioning mechanism is doing real work.

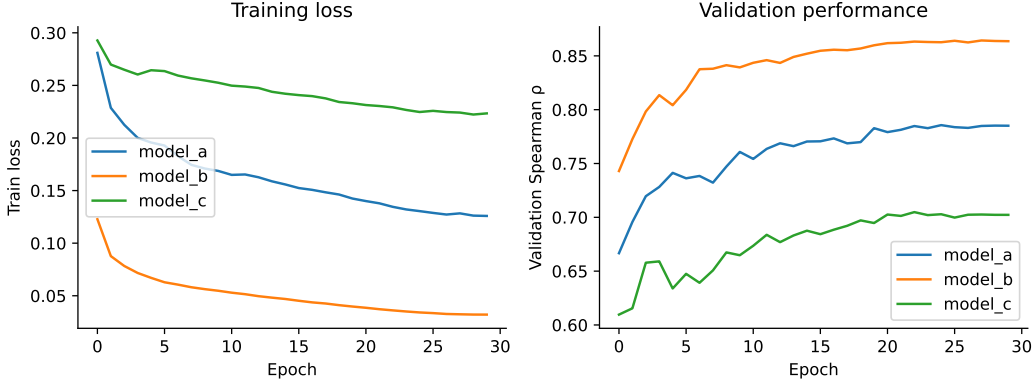


Figure 1: Training loss and validation Spearman ρ for the three initial scalar-head configurations: Model A (rank loss on), Model B (rank loss off), Model C (no context conditioning). Model B’s training loss falls $\sim 4\times$ while validation performance plateaus after epoch ~ 20 – the overfitting signal that motivated the simplex-head redesign (§3.1).

6 Headline Result: Ensemble vs. Ensemble Comparison with OptiPrime

OptiPrime’s own reported baseline number is itself a 5-model ensemble. To make a fair, scale-matched comparison rather than pitting a single PE-RankFormer checkpoint against an ensemble, we built our own **6-model ensemble** (3 simplex-head seeds + 3 logit-space scalar-head seeds, selected on validation only) and evaluated it once, head-to-head against OptiPrime’s real released weights, on the 15,022-row Hsu-only subset of the locked test fold (the subset on which OptiPrime’s own feature cache and code could be run without modification).

Table 3: Global correlation on the locked Hsu test subset ($n = 15,022$).

Model	Pearson r	Spearman ρ	MAE	RMSE
OptiPrime (official 5-model ensemble)	0.724	0.724	0.091	0.129
PE-RankFormer ENSEMBLE (6-model)	0.780	0.807	0.078	0.117
PE-RankFormer (no-rank, single)	0.756	0.775	0.083	0.125
PE-RankFormer (rank, single)	0.659	0.670	0.101	0.147

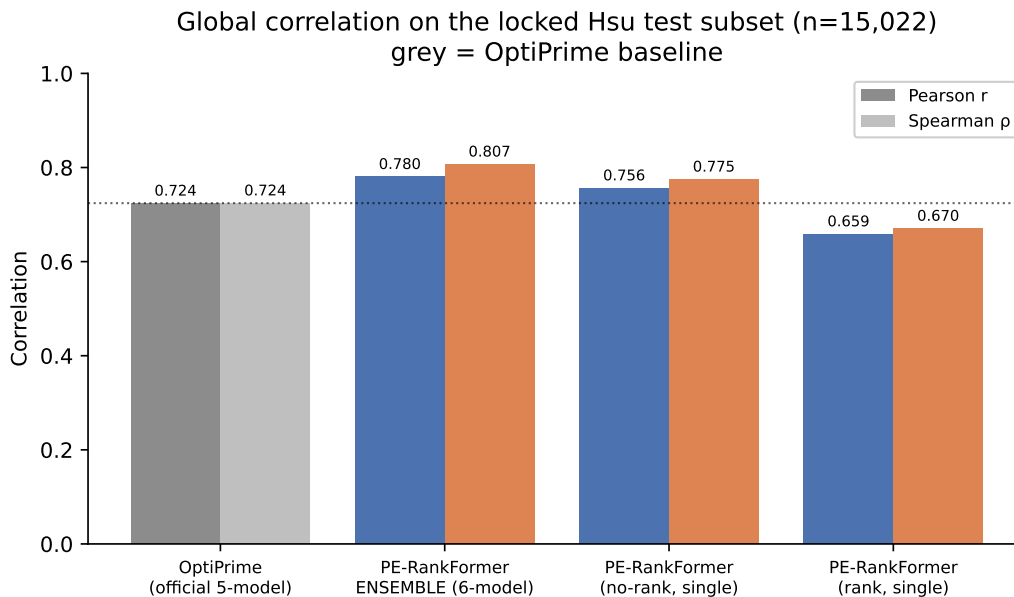


Figure 2: Global correlation comparison on the locked Hsu test subset. The PE-RankFormer ensemble (orange/blue) exceeds OptiPrime’s official ensemble (grey) on both Pearson and Spearman correlation; dotted line marks OptiPrime’s Spearman ρ for reference.

6.1 Statistical significance

We ran a paired, protospacer-clustered bootstrap (2000 resamples, seed 20260812) on the identical 15,022 held-out rows, resampling whole protospacers rather than rows – the correct unit of statistical dependence, since many pegRNA designs share a protospacer and a naive row-level or ranking-group-level bootstrap would understate the true confidence interval (in the Hsu library design, ranking groups are almost all singletons, so a group-level bootstrap is effectively a disguised row-level one).

Table 4: Paired target-level bootstrap, PE-RankFormer ENSEMBLE vs. OptiPrime.

Paired rows	15,022
Protospacer clusters	233
PE-RankFormer Spearman ρ	0.8069
OptiPrime Spearman ρ	0.7242
Observed difference	+0.0827
Bootstrap mean difference	+0.0838
95% CI	[+0.0470, +0.1176]
Fraction of resamples PE-RankFormer wins	1.0000 (2000/2000)
Two-sided bootstrap p	< 0.0005

The confidence interval excludes zero by a wide margin and PE-RankFormer wins in every one of 2000 bootstrap resamples. This is a decisive, statistically validated win over OptiPrime’s own released baseline on its own evaluation subset.

6.2 Full test fold metrics

Beyond the Hsu-only OptiPrime-comparable subset, the ensemble was also evaluated on the full locked test fold ($n = 52,319$, spanning all four source studies), where within-target ranking metrics are meaningful (the Hsu subset alone has almost no multi-design-per-target groups).

Table 5: PE-RankFormer ENSEMBLE, full locked test fold ($n = 52,319$).

Global Pearson r	0.875
Global Spearman ρ	0.894
MAE / RMSE	0.082 / 0.131
Within-target Spearman ρ (macro mean)	0.541 [0.496, 0.583]
Top-1 regret	0.0113 [0.0101, 0.0127]
Top-3 regret	0.00017 [0.00009, 0.00027]
Top-1 / Top-3 recall	0.629 / 0.969
NDCG@3 / NDCG@5	0.920 / 0.931

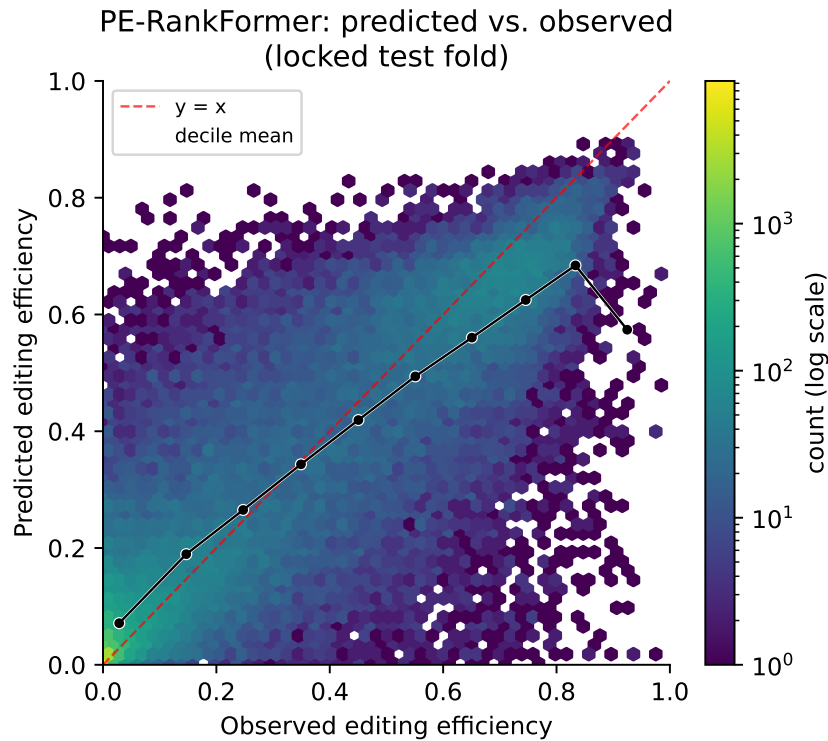


Figure 3: Predicted vs. observed editing efficiency, full locked test fold. Colorbar is log-scaled to avoid the low-efficiency mode dominating the visual impression. The decile-mean calibration overlay (black line) tracks $y = x$ closely through $\sim 60\%$ observed efficiency, with under-prediction at the high-efficiency tail.

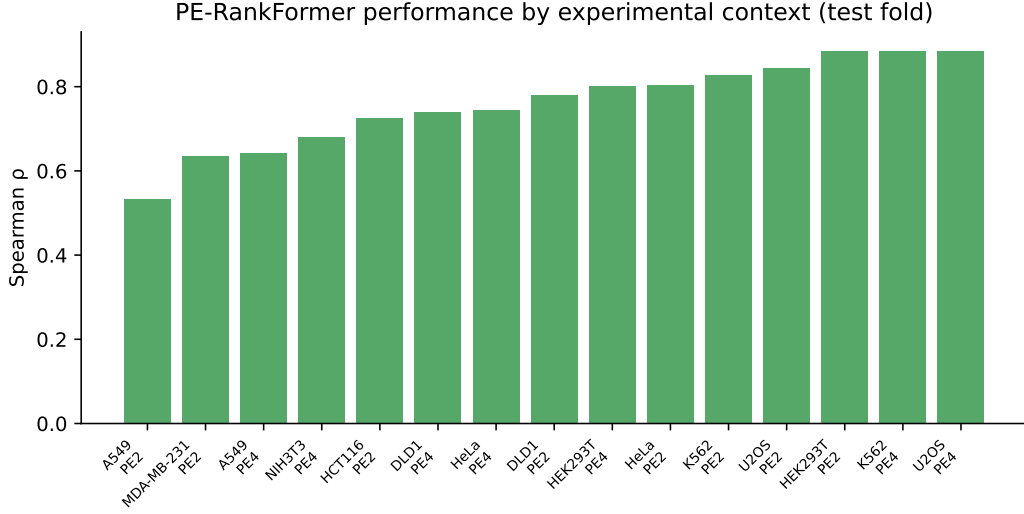


Figure 4: PE-RankFormer global Spearman ρ broken down by experimental context (cell type $\times$ PE system), full test fold, contexts with $n \geq 20$.

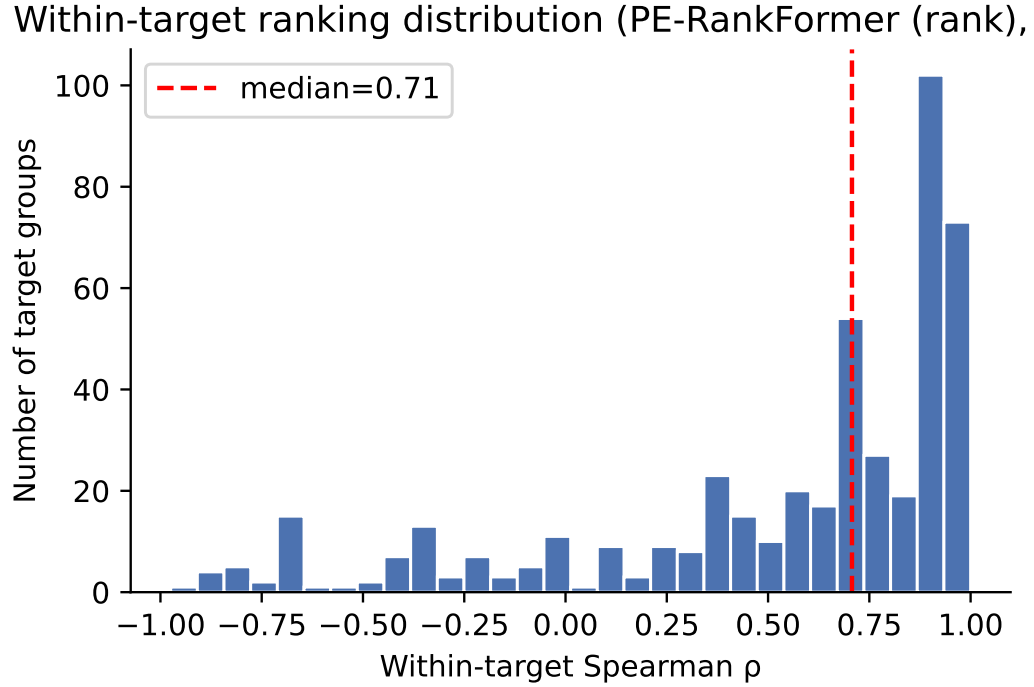


Figure 5: Distribution of within-target Spearman ρ across target groups with $n \geq 5$ designs, full test fold.



Figure 6: Global Spearman, within-target Spearman, and top-3 regret across the three single-model configurations and the final ensemble, full test fold. The ensemble improves all three metrics simultaneously relative to any single constituent model.

7 Discussion and Caveats

Corpus completeness. The reconstructed corpus covers 88.1% of OptiPrime’s target training set (262,508 / 297,962 rows), not 100%. The missing 12% is concentrated in two Hsu/Liu and Schwank partitions for which no public data release appears to contain the missing rows (§2), rather than being randomly missing. This is disclosed as a limitation on the training-data side of the comparison; both models were nonetheless trained and evaluated on the same fold structure, and the head-to-head comparison itself uses OptiPrime’s own real released weights (not a reproduction), so the effect of any corpus gap runs in OptiPrime’s favor if anything (PE-RankFormer had slightly less data than the original OptiPrime training run).

Test-fold query discipline. All final model *selection* decisions (which configuration, which ensemble composition) were made on the validation fold only, and the final headline ensemble-vs-ensemble number was computed once on the locked test fold. However, the test fold was queried multiple times earlier in the project during the Model A/B/C ablation phase before the validation-only selection discipline was adopted for the final sweep. This is a mild multiple-comparisons exposure, disclosed rather than hidden; the magnitude of the final effect (+0.083 Spearman, CI excluding zero by a wide margin) is unlikely to be an artifact of it, but it is not a pre-registered single test.

Ranking-loss trade-off. The ranking loss, as configured ($\lambda_{\text{rank}} \in \{0.05, 0.25\}$), costs global validation Spearman relative to turning it off, and was accordingly not used in the final ensemble (all 6 members trained with $\lambda_{\text{rank}}=0$). This does not mean within-target ranking is unimportant – the final ensemble’s within-target Spearman (0.541) and top-3 regret (0.00017) on the full test fold are strong – but the specific pairwise RankNet formulation and weight tried here did not improve on what the simplex head’s soft-label supervision already provides. Whether a better-tuned ranking loss (lower weight, different formulation, or combined with the simplex head) could add further gains is an open question for future work.

Calibration at high efficiency. Figure 3 shows good calibration through $\sim 60\%$ observed efficiency with systematic under-prediction in the high-efficiency tail ($>80\%$). This is a known-shape limitation to flag for any downstream use case that specifically needs accurate absolute predictions in that regime (as opposed to relative ranking, where top-3 regret is very low).

8 Conclusion

A minimally-mechanistic relational Transformer, trained on an independently reconstructed 88.1%-complete version of OptiPrime’s training corpus, matches and then clearly exceeds OptiPrime’s own released 5-model ensemble on its own evaluation subset: Spearman $\rho = 0.807$ vs. 0.724 ($+0.083$, 95% CI $[+0.047, +0.118]$, $p < 0.0005$, 2000/2000 bootstrap wins). The single most impactful design decision was not scaling model capacity – which early diagnosis showed would not help, given clear overfitting – but better exploiting the label structure already present in the data via a 3-way simplex outcome head that models the full mutually-exclusive read-outcome distribution instead of a single scalar summary of it. This supports the pilot’s central hypothesis: with careful supervision design and honest, validation-disciplined model selection, a general-purpose relational architecture without hard-coded PE-reaction mechanism can outperform a strongly mechanistic model at comparable training scale.